

Node Classification on Citation Networks using Graph Neural Networks with Logic Regularization

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1 Introduction and Project Goal

Many real-world datasets are naturally represented as graphs, where entities are modeled as nodes and relationships between them as edges. Examples include citation networks, social networks, and biological interaction networks. In such settings, observations are not independent, and classical machine learning methods that assume independent and identically distributed (i.i.d.) samples often fail to capture important structural dependencies.

Graph Neural Networks (GNNs) provide a powerful framework for learning from graph-structured data by explicitly modeling dependencies between connected nodes [2]. While GNNs have achieved strong performance on tasks such as node classification, standard approaches rely purely on data-driven likelihood maximization. They often ignore symbolic prior knowledge, which can be crucial when labeled data is scarce.

The goal of this project is to bridge the gap between deep learning and statistical relational learning. Specifically, I will investigate how incorporating a human-interpretable logical assumption—encoded as a **statistical prior**—affects node classification performance. Rather than enforcing hard logical constraints, symbolic knowledge is integrated as a differentiable regularization term, consistent with the probabilistic principles of statistical machine learning.

The main objectives of this project are:

- To implement a standard Graph Neural Network (GCN) as a probabilistic baseline.
- To formulate domain knowledge (homophily) as a logic-based regularization term.
- To test the hypothesis that logic regularization provides significant performance gains in **low-data regimes** (e.g., when only 1–5% of nodes are labeled), where the purely data-driven likelihood term is insufficient.
- To analyze how symbolic priors act as a smoothness constraint on the posterior label distribution.

This project aligns closely with the goals of CMPT 727 by emphasizing probabilistic modeling, structured dependencies, and the principled incorporation of prior knowledge into inference [3].

2 Dataset Description

2.1 Citation Network Dataset

The project uses the **Cora** citation network dataset, a widely used benchmark in statistical relational learning [4]. In this dataset:

- **Nodes** represent scientific publications.
- **Edges** represent citation relationships.
- **Features** are sparse bag-of-words vectors derived from paper abstracts.
- **Labels** correspond to one of seven research topics (e.g., Neural Networks, Genetic Algorithms).

2.2 Dataset Statistics

Table 1 summarizes the key statistics. The dataset is small enough for rapid iteration but complex enough to demonstrate non-trivial graph structure.

Table 1: Statistics of the Cora citation network dataset (Standard Planetoid Split).

Statistic	Value
Number of nodes	2,708
Number of edges (undirected)	5,278
Number of features per node	1,433
Number of classes	7
Average node degree	3.90

2.3 Exploratory Data Analysis

To understand the underlying structure, I performed an exploratory analysis. Figure 1 shows the class balance; while not perfectly uniform, it is balanced enough to avoid severe bias issues.

Crucially, Figure 2 (plotted on a log-log scale) reveals that the node degrees follow a heavy-tailed distribution. This power-law structure is characteristic of scale-free networks, suggesting that information propagation via GNNs will be heavily influenced by a few high-degree "hub" nodes.

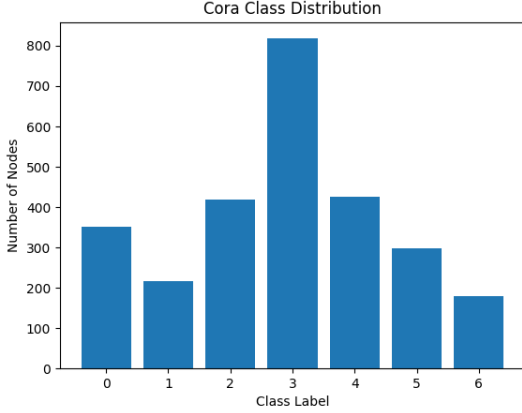


Figure 1: Class distribution of nodes in Cora.

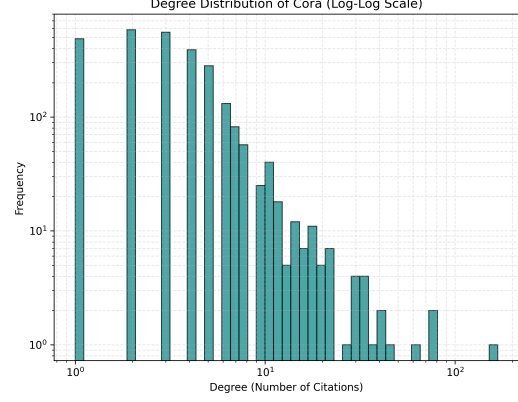


Figure 2: Node degree distribution (Log-Log Scale).

3 Methodology

3.1 Baseline: Graph Convolutional Network (GCN)

As a baseline, I will employ a Graph Convolutional Network (GCN) [2]. From a statistical perspective, the GCN parametrizes the conditional distribution $P(Y|X, A)$ of node labels. The propagation rule for layer l is:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)} \right) \quad (1)$$

where $\tilde{A}$ is the normalized adjacency matrix. The final output is a categorical distribution over the 7 classes, optimized by minimizing the cross-entropy loss $\mathcal{L}_{CE}$ (negative log-likelihood).

3.2 Logic-Regularized Extension

To incorporate symbolic knowledge, I will add a regularization term based on the **Homophily Assumption**: *Connected papers are likely to belong to the same research topic.*

Mathematically, I frame this not as a heuristic, but as a **Laplacian Prior** on the label distribution. This effectively models the citation network as a **Markov Random Field (MRF)**, encouraging connected nodes to have similar marginal probabilities.

For every edge $(i, j) \in E$, let p_i and p_j denote the predicted probability vectors. The logic regularization term is defined as:

$$\mathcal{L}_{\text{logic}} = \frac{1}{|E|} \sum_{(i,j) \in E} \|p_i - p_j\|_2^2 \quad (2)$$

The final training objective becomes a Maximum A Posteriori (MAP) estimation problem:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{\text{logic}} \quad (3)$$

where λ controls the strength of the prior. This approach represents a foundational instance of neuro-symbolic learning, where logical constraints guide the optimization of a neural function approximator [1].

3.3 Evaluation Protocol

Models will be evaluated using classification accuracy and macro-F1 score on a held-out test set (1,000 nodes). I will specifically compare the baseline GCN against the Logic-Regularized GCN across different values of λ and varying label rates (e.g., using only 20 labeled nodes vs. 140) to assess the robustness provided by the logic prior.

4 Expected Outcomes

This project is expected to demonstrate:

- A working implementation of a GNN pipeline in PyTorch Geometric.
- Empirical evidence that symbolic priors improve generalization, particularly when training data is sparse.
- Practical experience combining deep learning with probabilistic graphical model concepts.

This work will serve as a preliminary investigation for my master’s thesis on Neuro-Symbolic AI, establishing the software framework and theoretical understanding required for advanced Rule-Enhanced Graph Learning.

References

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- [2] Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations (ICLR)*, 2017.
- [3] Kevin P Murphy. *Probabilistic Machine Learning: An Introduction*. MIT Press, 2022.
- [4] Prithviraj Sen, Galileo Namata, Mustafa Bilgic, Lise Getoor, Brian Galligher, and Tina Eliassi-Rad. Collective classification in network data. *AI magazine*, 29(3):93–93, 2008.